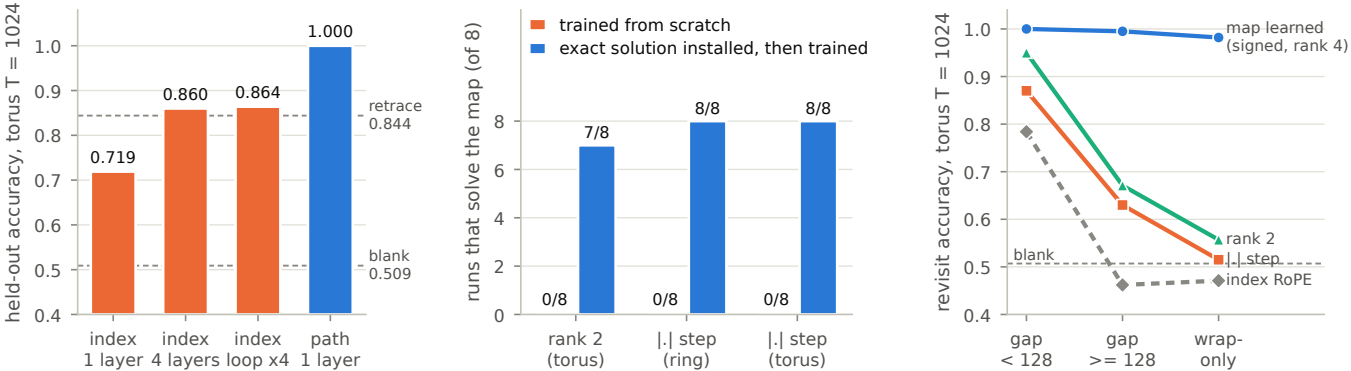
(a) One path layer learns the map; index depth does not
held-out accuracy, torus T = 1024
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.719
0.860
0.864
1.000
retrace 0.844
blank 0.509
index 1 layer
index 4 layers
index loop x4
path 1 layer
(b) The solution exists and training keeps or re-finds it, but training from scratch does not find it
runs that solve the map (of 8)
trained from scratch
exact solution installed, then trained
8
6
4
2
0
7/8
8/8
8/8
0/8
0/8
0/8
rank 2 (torus)
|.| step (ring)
|.| step (torus)
(c) Every failure keeps short returns and loses long and wrap-around ones
revisit accuracy, torus T = 1024
1.0
0.9
0.8
0.7
0.6
0.5
0.4
map learned (signed, rank 4)
rank 2
|.| step
index RoPE
blank
gap < 128
gap >= 128
wrap-only